

Progress Report – Phase 3

Project: Industry AI Flow – AI-Powered Construction Industry Platform

Program: Integrated Artificial Intelligence, SAIT Capstone Project

Team Members: - Angel Daniel Bustamante Perez - Jason Niu - Jack Si

Instructor: Reeta

Date: March 2026

Project Status Summary

Our system is feature-complete with all three core capabilities working: RAG Knowledge QA, Construction Cost Estimation, and Dynamic Data Analysis. We are now focused on ensuring demo stability, completing final testing, and preparing for the Capstone Showcase.

Task Distribution

Jason Niu – Software Development Lead

Task	Status	Description
System architecture design	Complete	Designed 6-layer architecture with 3 AI runtime paths
Backend API development	Complete	FastAPI with 10+ route modules, authentication, tenant isolation
RAG pipeline implementation	Complete	Hybrid search (BM25 + vector + RRF), BGE reranker, groundedness check
Intent classification system	Complete	11-node LangGraph StateGraph with multi-turn clarification
Cost estimation ML model	Complete	Ridge regression, 5-fold CV, $R^2 = 0.989$
Code execution sandbox	Complete	Docker-based isolated Python execution with security hardening
LLM integration (Ollama)	Complete	Qwen3.5:4b/9b with Metal GPU, hybrid local/cloud dispatch
Frontend development	Complete	Next.js App Router with chat, dashboard, cost estimation pages
Testing & quality gates	Complete	516+ unit tests, 70% coverage, 11-gate release validation
Performance optimization	Complete	Query caching, BM25 throttling, connection pooling
Documentation	Complete	Architecture docs, API reference, deployment guide

Jack Si – Construction Domain Expert

Task	Status	Description
Construction document collection	Complete	Curated ~12 industry documents (safety, codes, guides)
Cost estimation dataset	Complete	Provided 10,000-row construction project dataset
Domain knowledge validation	Complete	Verified RAG answers against industry knowledge
Feature selection for cost model	Complete	Identified key cost drivers: change orders, risk score, contractor rating
User acceptance testing	Complete	Tested system from construction professional perspective
Demo scenario preparation	In Progress	Preparing realistic demo queries and workflows

Angel Daniel Bustamante Perez – Project Support

Task	Status	Description
Project coordination	Complete	Team communication and milestone tracking
Document preparation	In Progress	Assignment submissions and presentation materials
Testing support	Complete	Assisted with system testing and bug reporting

Milestones

Milestone	Target Date	Status
Phase 1: Project proposal	Completed	Done
Phase 2: Data collection & preprocessing	Completed	Done
Phase 3: System architecture design	March 2026	Done
Core RAG pipeline functional	Completed	Done
Cost estimation model trained & integrated	Completed	Done
Dynamic data analysis with code sandbox	Completed	Done
Frontend UI complete	Completed	Done
36 rounds of test-driven improvement	Completed	Done
Demo preparation & rehearsal	Late March 2026	In Progress
Capstone Showcase	Late March / Early April 2026	Upcoming

Challenges and Solutions

Challenge	Solution
Ollama LLM performance slower than expected	We switched to the smaller Qwen3.5:4b model and made sure Metal GPU acceleration was enabled
PaddleOCR incompatible with Python 3.14	We locked our environment to Python 3.13.x and used PaddleOCR's nightly build
Intent classification getting stuck in loops	We added a <code>MAX_CLARIFICATION_ROUNDS = 2</code> limit to prevent infinite recursion
Code sandbox had security vulnerabilities	We ran 9 rounds of iterative security hardening to patch bypass issues
RAG queries were too slow for demo	We added response caching, throttled BM25 index rebuilds, and reused TCP connections

Next Steps

1. Final demo rehearsal with realistic construction scenarios
2. Prepare presentation slides for Capstone Showcase
3. Ensure system stability under sustained demo usage
4. Complete remaining assignment submissions